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Cohesin reconstitution and homologous recombination repair of DNA double strand breaks in late mitosis

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eLife Assessment

This study provides **convincing** evidence that homologous recombination can occur in telophase-arrested cells, independently of cohesin subunits Smc 1-3. These findings are **valuable** as they point to investigate the role of cohesins re-association with chromatin in the allelic inter-sister repair by homologous recombination.

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Abstract

The cohesin complex maintains sister chromatid cohesion from S phase to anaphase onset. Cohesin also plays roles in chromosome structure and DNA repair. At anaphase onset, the cohesin subunit Scc1 is cleaved to allow segregation in an orderly manner, although some residual cohesin subunits remain to maintain chromosome structure. Efficient DNA double strand break (DSB) repair by homologous recombination (HR) with the sister chromatid also depends on cohesin. Here, we have examined the role of residual cohesin in DSB repair in telophase (late mitosis). We have found that Scc1 returns in telophase after DSBs, and that it partially reconstitutes a chromatin-bound cohesin complex with Smc1 and an acetylated pool of Smc3 after a single HO-induced DSB at the *MAT* locus. However, this new cohesin is neither required for the HR-driven *MAT* switching nor binds to the *MAT* locus after the DSB.

Introduction

Cell survival and genome integrity depend on the faithful segregation of chromosomes in anaphase. From the S phase until the anaphase onset, the highly conserved cohesin complex plays an essential role in the structural maintenance of sister chromatids. Cohesin is a multiprotein complex composed in yeast of Smc1, Smc3, Scc1 (also known as Mcd1), Scc3 and Pds5 subunits (Hirano, 2000; Michaelis et al., 1997; Sjögren and Nasmyth, 2001). At the core of the complex, Smc1, Smc3 and Scc1 form a heterotrimeric ring that embraces and holds the replicated sister chromatids together and well-aligned until reaching G_2/M (Díaz-Martínez et al., 2008; Koshland and Guacci, 2000; Laloraya et al., 2000; Michaelis et al., 1997; Strunnikov et al., 1993). Cohesin subunits are loaded onto chromosomes in G1 by the Scc2-Scc3 complex (Ciosk et al., 2000). When cells enter S phase, the acetyl-transferase Eco1 acetylates Smc3, which inhibits the ATPase activity of Smc1-Smc3 heads and prevents the opening of the Smc3-Scc1 interface (Çamdere et al., 2015; Chan et al., 2012; Huber et al., 2016; Murayama and Uhlmann, 2015; Ström et al., 2007; Unal et al., 2007). This embraces the nascent sister chromatids and establishes cohesion (Uhlmann and Nasmyth, 1998). Once sister chromatids are ready for segregation, the APC^{Cdc20} (Anaphase Promoting Complex associated with its regulatory cofactor Cdc20) initiates anaphase by degrading securin/Pds1, so that the separase/Esp1 becomes active. Separase then cleaves the Scc1 subunit by proteolysis, releasing sister chromatids from cohesion (Cohen-Fix et al., 1996; Michaelis et al., 1997; Nasmyth and Haering, 2009; Uhlmann et al., 1999; Yamamoto et al., 1996). The fragmented Scc1 is then rapidly degraded (Rao et al., 2001), while a pool of Smc1-Smc3 dimers appears to remain through anaphase (Renshaw et al., 2010; Tanaka et al., 1999). This pool however becomes loose as Smc3 is deacetylated by Hos1 in anaphase (Chan et al., 2012; Huber et al., 2016; Murayama and Uhlmann, 2015). Interestingly, some cohesin-dependent cohesion remains at chromosome arms during anaphase, suggesting that residual cohesin is present despite separase activation (Djeghloul et al., 2020; Garcia-Luis et al., 2022; Renshaw et al., 2010).

In addition to its role in sister chromatid cohesion, cohesin has also been involved in chromosome structure and DNA repair. On the one hand, cohesin can hold two DNA segments within the same chromatin forming an extruded loop (Lazar-Stefanita et al., 2017; Rao et al., 2017; Schalbetter et al., 2017). On the other hand, DNA double strand breaks (DSBs) produce the so-called damage-induced cohesion (DI-cohesion) (Kim et al., 2010).

Following DNA damage, cohesin is recruited and accumulated both along 50 – 100 kb surrounding the DSB site and genome-wide (Ström et al., 2004; Ünal et al., 2004). The cohesin recruitment creates a firm anchoring of two well-aligned sister chromatids, which in turn facilitates DSB repair by homologous recombination (HR) (Hou et al., 2022; Phipps and Dubrana, 2022). In this regard, DNA damage kinases such as Mec1 and Chk1 phosphorylate Scc1 to initiate the cohesion establishment in post-replicative cells (Heidinger-Pauli et al., 2008).

HR is a DSB repair mechanism that uses a homologous template for restoring the original broken DNA. HR is highly reliable when the correct template is used, which in mitotic cells is the sister chromatid. Hence, HR is the preferred repair mechanism from S to G2/M, when the sister chromatid is available in close proximity (Langerak and Russell, 2011; Mathiasen and Lisby, 2014; Symington et al., 2014). Before DNA replication, in G1, the alternative error-prone non-homologous end joining (NHEJ) is used instead. Cells coordinate the choice of either NHEJ or HR based on the activity of cyclin dependent kinase (CDK). Low CDK keeps cells in G1 and favours NHEJ, whereas high CDK is present in S and G2/M and activates HR. Noticeably, there is a paradox in this link between high CDK and HR. In the cell cycle window that spans from anaphase to the telophase-to-G_1 transition (we will refer to this window as late mitosis), high CDK is set to favour HR despite a sister chromatid is not in proximity (Machín and Ayra-Plasencia, 2020). In a previous study, we showed that sister loci can indeed move closer and coalesce, and that HR still appears important for DSB survival in late mitosis (Ayra-Plasencia and Machín, 2019). Considering all these observations, we wondered whether the residual cohesin is affected by DSBs in late mitosis (telophase) and whether this cohesin pool is important for HR repair.

Results and Discussion

Scc1 becomes stable after DSBs in late mitosis

At the anaphase onset, Esp1 cleaves Scc1, opening the cohesin ring and releasing sister chromatids from cohesion. Thereafter, Scc1 needs to be translated *de novo* since cleaved fragments are unstable and degraded by the proteasome (Rao et al., 2001). However, it is known that activation of Esp1 can be blocked by DDC kinases (Yam et al., 2020). In addition, DDC kinases also phosphorylate Scc1 to facilitate DI-cohesion at a post-replicative stage (Heidinger-Pauli et al., 2008). Thus, it is feasible that DSB generation in late mitosis can stabilize *de novo* Scc1 as well as rendering it cohesive. Noteworthy, Smc1 and Smc3 subunits remain attached to chromatin after segregation (Garcia-Luis et al., 2022; Renshaw et al., 2010; Tanaka et al., 1999).

To assess how cohesin subunits behave after DSBs in late mitosis, *cdc15-2* strains that bear Scc1 tagged with myc epitopes and Smc1/3 tagged with HA epitopes were first blocked in telophase at 34 ºC for 3 hours (Tel) (**Fig. 1a**). The *cdc15-2* allele encodes a conditionally thermosensitive Cdc15 kinase, which drives telophase-to-G1 transition (Machín and Ayra-Plasencia, 2020). These strains also harbour an inducible HO endonuclease, which cuts the HO cutting site (*HOcs*) at the *MAT* locus in chromosome III. The HO gene is under the control of a promoter systemthat responds to β-estradiol (*lexO4p:HO*:plus LexA-ER-B112-T) (Gnügge and Symington, 2020; Ottoz et al., 2014). In wild type strains, the HO DSB is repaired by HR with the intramolecular ectopic loci *HMR* and *HML*. The locus election depends on the *MAT* allele (the *MAT*a HO DSB repairs with *HML*) and results in a gene conversion (e.g., from *MAT*a to *MAT*α). In order to undoubtedly attribute the effects on cohesin to the DSB and not to downstream events of the repair process, we used a set of strains unable to repair the HO-mediated DSB (Δ*hml* Δ*hmr* double mutant). After the Tel arrest, the cultures were divided into three. The first subculture served as a mock control, the second was used to generate the single HO DSB (+HO), and the third was used to generate multiple random DSBs by means of the radiomimetic drug phleomycin (+phle). Noteworthy, these latter DSBs are repairable in principle. Incubation of the subcultures was prolonged at 34 ºC for 2 extra hours.

The presence or absence of Scc1-3myc, Smc1-6HA, and Smc3-6HA was checked by Western blot. As controls, we included asynchronous samples from the same strains, before the telophase arrest, as well as a parallel arrest at G2/M by the drug nocodazole (Nz). As expected, Smc1, Smc3 and Scc1 bands were detected in both cycling (asyn) cells and cells blocked in G2/M (**Fig. 1b-d**). The Scc1 signal disappeared almost entirely when cells were arrested in telophase and remained low in the mock control two hours later. Strikingly, full-length Scc1 was restored after single and multiple DSBs generation (**Fig. 1b,e**).

Compared to the mock condition, phleomycin treatment increased Scc1 levels fivefold, reaching approximately half of the levels observed in G2/M. In the case of HO DSBs, Scc1 levels surpassed those of the mock by more than fifteenfold and were even higher than in G2/M. In contrast, Smc1 and Smc3 levels remained more stable in both types of DSBs (**Fig. 1b-d**).

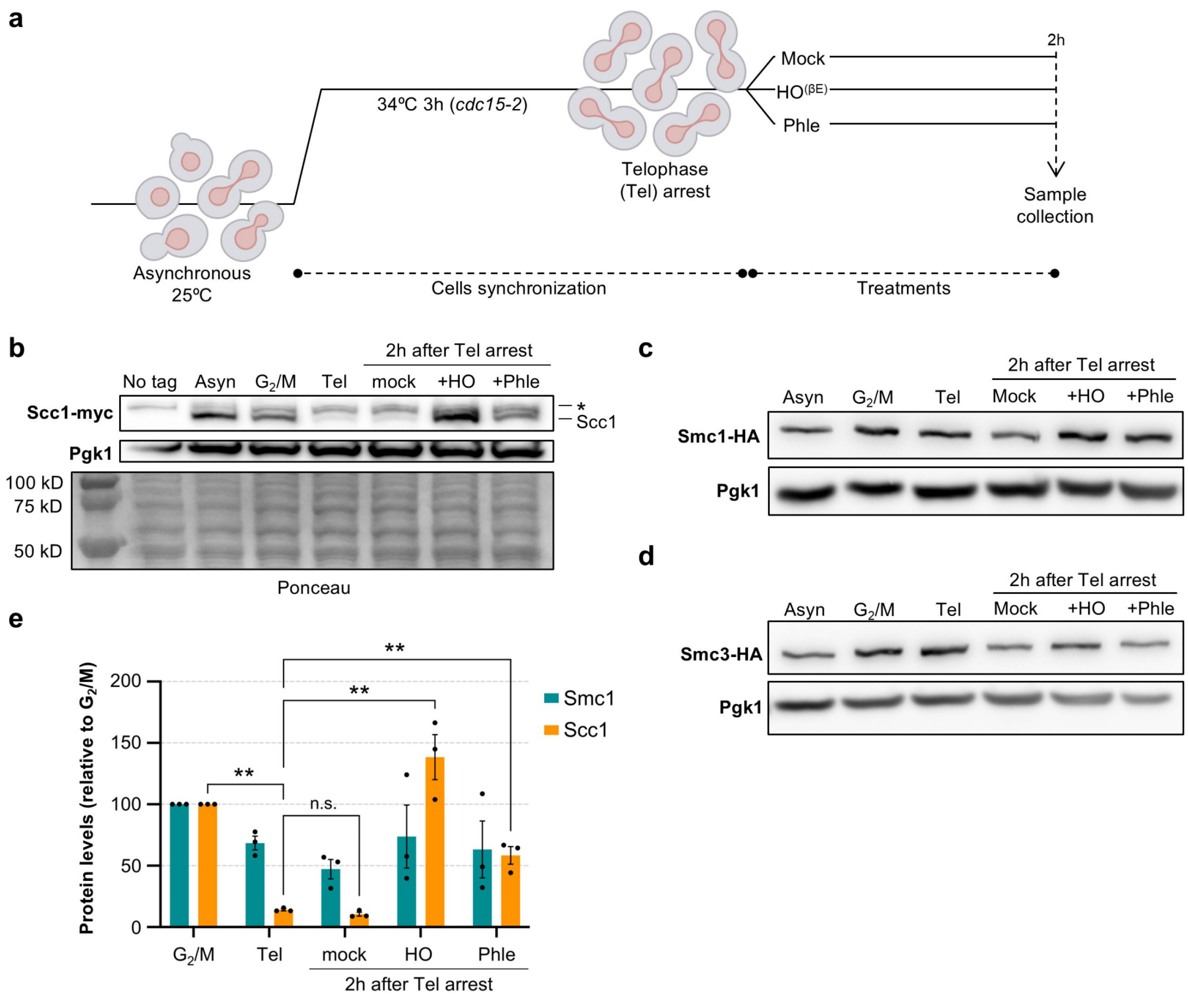


Figure 1.

Scc1 returns after DSBs in late mitosis.

(**a**) Schematic of the experimental procedure. Cells in logarithmic growth phase were first arrested in telophase (Tel) by incubating them at 34ºC for 3 h (strains bear the *cdc15-2* allele). Then, the culture was divided into three subcultures. One served as a mock control, and the other two were treated to generate DSBs, one with β-estradiol (to express the HO endonuclease) and the other with phleomycin. The strain is unable to repair the HO-DSB break by HR-driven gene conversion (Δ*hmr* Δ*hml*). (**b**) Western blot against Scc1-3myc. This experiment compares Scc1-3myc levels in an asynchronous, G2- and telophase-blocked cultures. The leftmost lane (No tag) is a control strain for Scc1 without the 3myc epitope tag. Pgk1 protein served as a housekeeping. Ponceau S staining is also shown as a loading control. Asyn.: Asynchronous. Tel: Telophase. +Phle: 10 mg·mL^{-1} phleomycin. +HO: 2 μM β-estradiol. *: Unspecific band detected by the α-myc antibody just over the Scc1-3myc signal. (**c**) Like in (b) but against Smc1-6HA. (**d**) Like in (c) but against Smc3-6HA. (**e**) Quantification of Scc1-3myc and Smc1-6HA levels in all conditions (mean ± sem, n=3). Statistical comparisons are shown for selected pairs (**, p<0.01; one-way ANOVA, Tukey's post hoc).

The cohesin complex is reconstituted and binds to chromatin after the HO DSBs in telophase

The above findings suggest that the cohesin complex could be newly formed and loaded onto the breaks, promoting DI-cohesion, which could in turn ease the recombinational events. To assess whether cohesin subunits are reassembled after DSB induction in telophase, we performed a co-immunoprecipitation (co-IP) experiment using Smc1-6HA as the bait protein in a strain that also expresses Scc1-3myc (**Fig. 2a** and **S1**). In our experimental setup, in addition to the anti-HA and anti-myc antibodies, we included an antibody for the acetylated form of Smc3 at lysine residues K112 and K113 (acSmc3), which have been linked to the chromatin-bound pool of the Smc1-Smc3 dimer (Beckouët et al., 2010; Minamino et al., 2015; Rolef Ben-Shahar et al., 2008; Unal et al., 2008). In asynchronous cultures, both Scc1-3myc and acSmc3 co-immunoprecipitated with Smc1-6HA, indicating that we were able to pull down the chromatin-bound pool (acetylated Smc3) of the canonical Smc1-Smc3-Scc1 complex. A parallel co-IP using untagged Smc1 did not recover the other subunits, confirming the specificity of the interaction. As expected, Scc1 was not detected in the pull-down under the mock condition (no DSBs); however, it was recovered after generation of the HO DSB. Surprisingly, this was not the case when random DSBs were induced by phleomycin. Incidentally, acSmc3 appeared as two bands in the asynchronous culture but as a single upper band in all telophase conditions, suggesting a difference in posttranslational modifications at this cell cycle stage.

The reconstitution of the complex and the detection of the acetylated form of Smc3, prompted us to confirm whether the newly formed Scc1 was bound to chromatin, since the coexistence of two pools could not be ruled out based on the co-IP experiments alone. For instance, a soluble Smc1-Smc3-Scc1 complex could coexist with a chromatin-bound Smc1-acSmc3 dimer, which could represent up to 30% of the total Smc1-Smc3 in anaphase (Beckouët et al., 2010). To this end, we performed biochemical fractionation of the chromatin-bound proteins, observing an increase in all cohesin subunits in the chromatin fraction after the HO DSB (**Fig. 2b** and **S2**). The increase was approximately twofold for Smc1-acSmc3 and up to sevenfold for Scc1. These results confirmed that the new Scc1 binds to a reconstituted chromatin-bound cohesin complex following the HO DSB. In contrast, this chromatin recruitment was not observed after phleomycin treatment. The differences in co-IP and fractionation due to the source and nature of the DSBs are intriguing. In this experimental setup, the HO DSB is irreparable, whereas DSBs caused by phleomycin can eventually be repaired by anaphase regression and sister loci coalescence (Ayra-Plasencia and Machín, 2019). In addition, the HO DSB leaves clean ends that can be easily resected, while the DSBs after phleomycin are chemically modified and require extended processing.

Next, we used chromatin immunoprecipitation (ChIP) to investigate whether the HO DSB recruits more cohesin to the vicinity of the *HOcs*. This recruitment has been reported during DSBs that exert a G2/M arrest by the DDC (Ström et al., 2004), Because the level of Scc1 changes so dramatically after the HO DSB, we selected one of the subunits that does not for the ChIP. An Smc3-HA strain unable to repair the HO DSB (Δ*hml* Δ*hmr* double mutant) was arrested in both G2/M and telophase. We confirmed an increase in the Smc3-bound pool around the *HOcs* in G2/M after the DSB (**Fig. S3**), however such an increase was not observed in telophase (**Fig. 2c**). We thus conclude that the core of the cohesin SMCs (Smc1-Smc3) are not further recruited to the HO DSB in late mitosis. This lack of local recruitment could be due to several non-exclusive scenarios, including the inability to increase the acSmc3 pools by Eco1, whose levels drop after G2/M (Lyons and Morgan, 2011), as well as condensin binding to chromosome arms during anaphase (Leonard et al., 2015), which might act as an eviction barrier for cohesin loading. However, reassembly of the full cohesin complex at residual Smc1-acSmc3 dimers may shift the organization of the surrounding chromosome regions and impinges on the HO DSB repair in telophase.

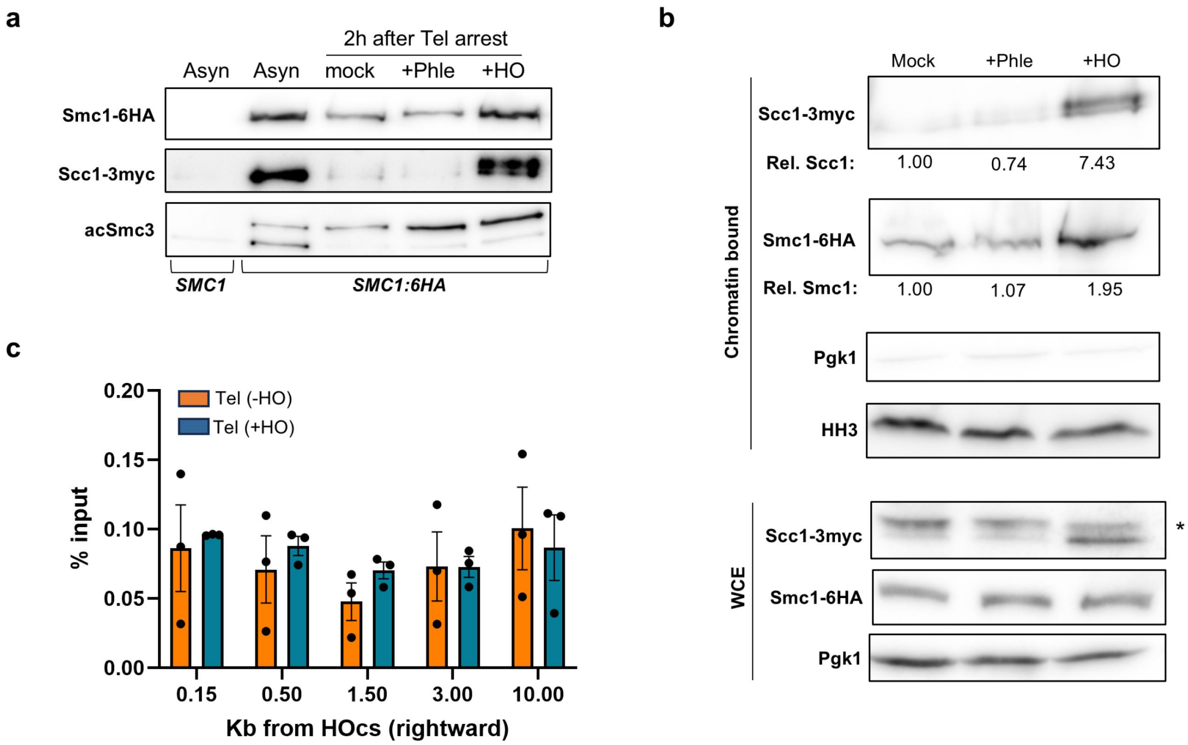


Figure 2.

Scc1 forms a reconstituted chromatin-bound cohesin complex after the HO DSB in late mitosis.

Cells were treated as in **Figure 1a**. (**a**) Co-IP of the core cohesin complex. Smc1-6HA was used as the bait protein; the leftmost lane includes a control with an untagged Smc1. (**b**) Chromatin fractionation of the strain expressing Smc1-6HA and Scc1-3myc. Relative protein levels (to the mock sample) are indicated under the blots. Pgk1 and histone H3 (HH3) were included as reporters of the cytosolic and chromatin-bound fractions, respectively. The asterisk (*) indicates an unspecific band in the whole cell extract (WCE), that is absent from the chromatin-bound fraction. (**c**) Cohesin does not bind the HO DSB in telophase. Chromatin immunoprecipitation (ChIP) of Smc3-6HA with (+HO) and without (-HO) HO induction (mean ± sem, n=3).

The HO-mediated DSB is efficiently repaired by HR in late mitosis

In a previous report, we showed from both genetic and cytological points of view that HR was still active in late mitosis (Ayra-Plasencia and Machín, 2019). Having observed that cohesin is reconstituted and chromatin-bound after the HO DSB in telophase, we next wondered about the functional role of the complex in this context.

We first tested whether the HO-induced DSB is efficiently resected into 3' ssDNA tails in late imtosis. Resection is necessary to formthe protruding 3' Rad51 nucleoprotein filaments that invade the donor homologous sequence to restore the break by HR (Peng et al., 2021; Symington, 2016). To confirm and quantitate the formation of ssDNA flanking the *HOcs*, we used a qPCR approach whereby primers can only amplify a target DNA cut with the restriction enzyme *StyI* if the target has been rendered single-stranded by resection (**Fig. S4**) (Gnügge et al., 2018; Zierhut and Diffley, 2008). We used the Δ*hml* Δ*hmr* double mutant to hamper the normal HR flow from resection to invasion and thus boosts ssDNA detection. Two different locations downstream of the *HOcs* were monitored to determine both the initial resection (0.7 Kb from *HOcs*) and the kinetics further away (5.7 Kb). In addition, cells arrested in either G2/M or telophase were directly compared to each other. Cells were first arrested for 3 h, the HO was induced afterwards, and samples were taken every hour during a 4 h time course while sustaining the HO induction (**Fig. 3a**). As expected, cells in G2/M efficiently resected the *HOcs* ends. The time to reach half of the maximum resection ($t_{1/2}$) was ~1 h at 0.7 Kb and ~2.5 h at 5.7 Kb from the DSB, respectively. Resection in telophase was only slightly delayed at 0.7 Kb ($t_{1/2}$ ~1.5 h), whereas the delay was more pronounced at 5.7 Kb ($t_{1/2}$ ~ 3.5 h). The delay may be due to either the fact that CDK activity in late mitosis, though still high, is slightly inferior than that at G2/M or the different chromosome compaction degrees between the arrests. Together with the samples for the qPCR, samples for Western blot of Rad53 were taken. Rad53 is an effector kinase of the DDC that gets activated by hyperphosphorylation after being recruited to the 3' ssDNA overhands (Branzei and Foiani, 2006). In agreement with the resection profile, Rad53 hyperphosphorylation occurred in late mitosis, although with ~1 h delay (**Fig. 3b**). This delay is also consistent with previous studies on Rad53 activation upon an unrepairable HO DSB (Pellicioli et al., 2001).

Once resection begins, cells are normally committed to recombine because long ssDNA fragments do not work as efficient substrates for NHEJ. In our haploid strain, HR at the *MAT* locus after the HO-driven DSB implies the gene conversion from the *MAT*a to the *MAT*α allele, whereas repair by NHEJ results in reconstitution of the *MAT*a allele (**Fig. 3c**) (Yamaguchi and Haber, 2021). This gene conversion can be detected by Southern blot because of a restriction fragment length polymorphism for *StyI*. Cells that are in G1 can barely repair the HO DSB, whereas those in G2/M do so through HR (Ira et al., 2004). To check whether HR was efficient at the HO DSB in late mitosis, we performed an experimental setup similar to the one shown above, but with two important differences. On the one hand, the strain retains the *HML* locus, enabling HR if engaged, and on the other hand, the HO endonuclease was transiently induced by 1 h β-estradiol pulse, allowing time for the complete repair of the induced DSB. Samples were taken at the time of the arrest, after the pulse, and all through the repair window. We confirmed by Western blot that HO (HO tagged with the Flag epitope) is produced after the pulse, and is rapidly degraded afterwards (**Fig. 3d**, HO-Flag blot). Rad53 hyperphosphorylation followed the ~1 h delay shown above, and became prominent by the time the DSB inducer had been removed (**Fig. 3d**, Rad53 blot), suggesting extensive resection even in this repairable setup. The Southern blot showed that (i) most *MAT*a locus in the cell population has been efficiently cut after the HO pulse (**Fig. 3e** and **f**; decrease of the *MAT*a band intensity and rise of that of the cut *HOcs*); and (ii) most DSBs are repaired afterwards exclusively through HR (**Fig. 3e** and **f**; drop of the HO cut band intensity and rise of that of the *MAT*α). No signs of DSB repair by NHEJ were observed (the remaining *MAT*a band stays constant throughout). This was further confirmed by measuring the repair kinetics of a derivative strain deficient in NHEJ (*yku70*Δ; **Fig. S2a-c**), and is consistent with our previous

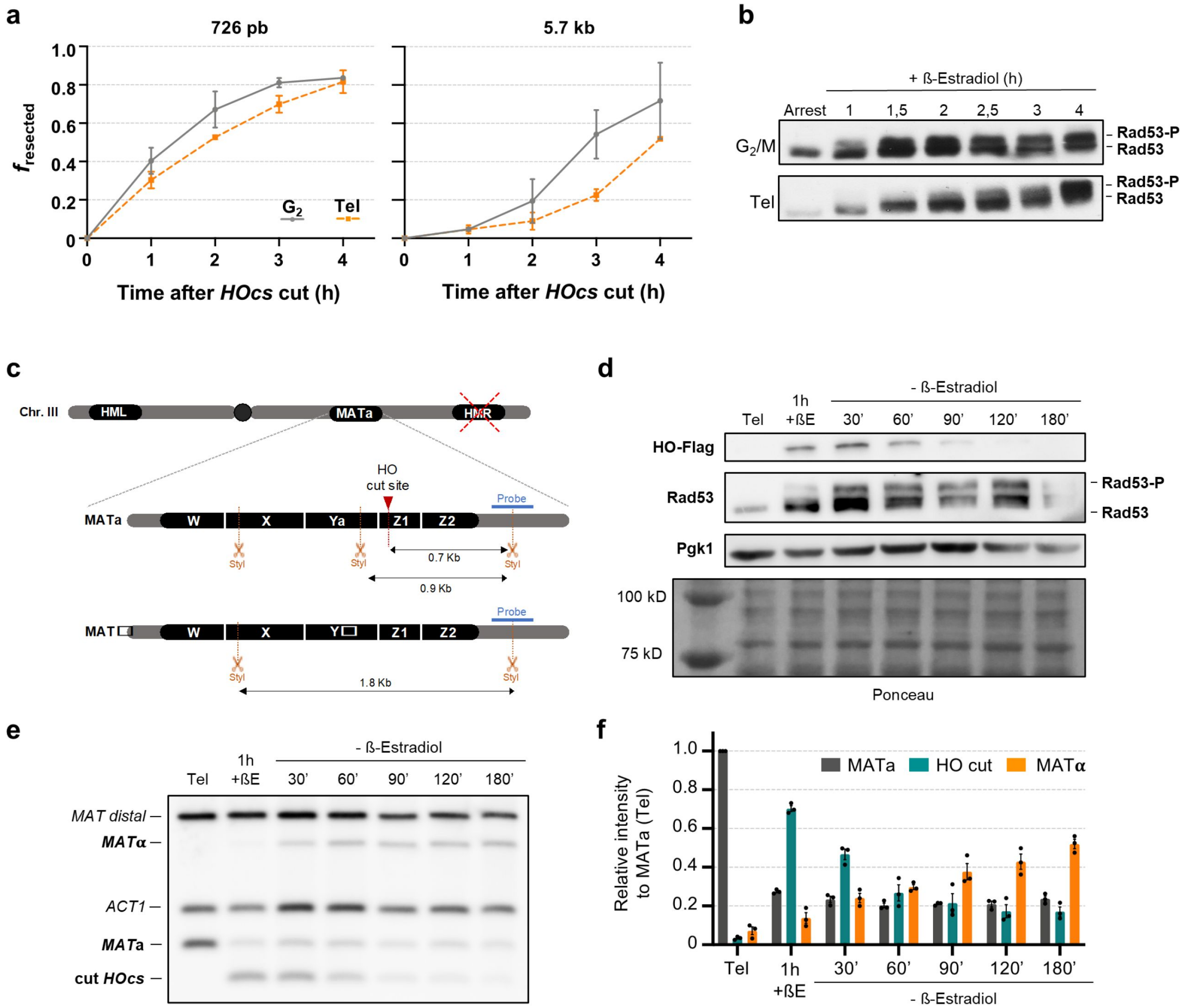


Figure 3.

Homologous recombination repairs the HO-DSB in telophase.

(**a**,**b**) Resection in late mitosis is almost as efficient as in G2/M. Cells were arrested in G2/M or telophase (Tel), and then the HO DSB was generated by adding β-estradiol. Samples were taken at the indicated times to monitor the kinetics of resection. This strain is unable to repair the HO-DSB break by HR-driven gene conversion (Δ*hmr* Δ*hml*). (**a**) Charts depicting the resection kinetics for two different amplicons located at 726 bp and 5.7 kb downstream the *HOcs* break (mean ± sem, n=3; $f_{resected}$ is the proportion of resected DNA). (**b**) Representative western blots against Rad53 to follow the sensing of DNA damage detection in G2/M versus late mitosis. (**c-f**) Yeast cells use HR to repair the HO DSB in late mitosis. Cells were first blocked in telophase (Tel). Then, the HO DSB was generated by adding β-estradiol. After 1 hour, the β-estradiol was washed away, and samples were taken to monitor the repair for 3 hours. The strain can repair the HO-DSB break by HR-driven gene conversion (Δ*hmr HML*). (**c**) Schematic of the fragments obtained after a StyI digestion for both *MAT*a and *MAT*α sequences. The probe to detect the fragments by Southern blot is in blue. When the *MAT*a locus is intact, the digestion gives rise to a 0.9 kb fragment. The HO cutting site (*HOcs*) is located within the StyI-digested *MAT*a locus. Thus, the HO-driven DSB shortens the fragment to 0.7 kb. HR leads to a gene conversion to *MAT*α, which results in the loss of a *StyI* restriction site and a new fragment of 1.8 kb. (**d**) Representative Western blot analyses for HO induction and subsequent degradation (tagged with Flag epitope) and the DSB sensing through Rad53 hyperphosphorylation. A Pgk1 western blot served as a housekeeping, and Ponceau S staining of the membrane as a loading control for all lanes. The leftmost lane in the Ponceau S corresponds to the protein weight marker. (**e**) Representative Southern blot for the *MAT* switching assay in late mitosis. Alongside the *MAT* probe, a second probe against the *ACT1* gene (1.1 kb fragment) was included for normalization. The *MAT* probe also recognizes an allele-independent *MAT* distal fragment (2.2 kb). (**f**) Quantification of relative band intensities in the *MAT* switching Southern blots (mean ± sem, n=3). Individual values were normalized to the *ACT1* signals. Then, every lane was normalized to *MAT*a at the arrest. Tel: Telophase. +βE: β-Estradiol addition.

studies with a HR-deficient *rad52*Δ strain (Medina-Suárez et al., 2024). Interestingly, HR kinetics were not altered in cells unable to activate the DDC (*rad9*Δ; **Fig. S2d-f**) or to tether the DSB ends (*mre11*Δ; **Fig. S2g-i**).

These results are also consistent with previous studies in G2/M cells, in which HR-driven repair of the HO-DSB appeared more robust than other types of DSBs (Lazzaro et al., 2008; Moreau et al., 2001).

Overall, we concluded that HR is the chosen repair pathway in late mitosis. This is in agreement with predictions based on CDK activity and also fits well with the genetic and cytological data we have obtained before (Ayra-Plasencia and Machín, 2019; Machín and Ayra-Plasencia, 2020; Medina-Suárez et al., 2024). By contrast, this emphasizes the paradoxes of having HR in the context of mostly segregated sister chromatids and without the cohesin complex holding them together as expected. We have now explored the second paradox further as we have just shown that cohesin is reconstituted after DSBs in telophase, and DI-cohesion appears important for efficient DSB repair by HR (Hou et al., 2022; Phipps and Dubrana, 2022).

The MAT switching in late mitosis does not depend on cohesin

In cells arrested in *cdc15-2*, the Smc1-Smc3 pool mostly resides at centromeres where it brings together adjacent regions in both chromosome arms (Garcia-Luis et al., 2022). Of note, this configuration could favour the physical interaction in *cis* of the *MAT* and the *HML* loci (Piazza et al., 2021), which may in turn promote a rapid and efficient HR-driven *MAT* switching. Thus, we tested whether *MAT* switching was dependent on this residual centromeric cohesin. To this aim, we tagged the Smc3 subunit with the auxin-mediated degron system (aid*). To check that the system was working correctly, a serial dilution spot assay was carried out (**Fig. 4a**). The strain expressing Smc3-aid* was plated on YPDA and YPDA plus 8 mM of the auxin indole-acetic acid (IAA). Alongside, we also plated a derivative strain where we had removed the preceptive F-box protein OsTIR1, which forms the functional ubiquitin ligase responsible for targeting the aid* for degradation upon IAA addition (Morawska and Ulrich, 2013). Since cohesin is an essential complex for vegetative growth, the lack of growth in the *SMC3-AID* OsTIR1* strain strongly points out that Smc3-aid* is degraded efficiently.

Next, we used this strain to degrade Smc3 after cells have been arrested in late mitosis and before the HO DSB. After IAA addition, there was a clear decline in Smc3-aid* levels, including the chromatin-bound acSmc3-aid* pool (**Fig. S3a**). After Smc3 depletion, a 1 h pulse of HO expression was applied, which allowed us to follow both DSB generation and its subsequent repair without residual cohesin. More than 90% of the Smc3-aid* had disappeared by the time the HO promoter was shut down (**Fig. S3a**). The addition of IAA drastically delayed gene conversion from *MAT*a to *MAT*α (**Fig. S3b,c**), although the cut efficiency was lower than in the wild type in the absence of auxin (~ 50 % *MAT*a remained uncut in the *SMC3:aid* OsTIR1* strain). Also, the detection of DNA damage through Rad53 phosphorylation seems fainter and slightly delayed (**Fig. S3a**). To unequivocally determine whether *MAT* switching kinetics were affected by Smc3 depletion, and not by an off-target effect of IAA, we compared strains that could either degrade or retain Smc3 in the presence of IAA. Four isogenic strains were analysed, comprising all possible combinations of Smc3-aid* and OsTir1, as confirmed by Western blot (**Fig. 4b**; Tel samples). The addition of IAA induced Smc3-aid* degradation only in the strain co-expressing OsTIR1 (**Fig. 4b**; +IAA samples). In the wild type strain (no tagged Smc3 and no OsTIR1), the cutting efficiency of *HOcs* and the yield of gene conversion to *MAT*α were both reduced in the presence of IAA, confirming that IAA itself negatively impacts *MAT* switching (**Fig. 4c**, compared two leftmost lanes with **Fig. 3e**). However, when the yield of the *MAT* conversion after the recovery time was normalized to the cut *HOcs* when HO was repressed, no significant differences were observed among the four strains (**Fig. 4c,d**). We therefore conclude that the core of the cohesin complex is

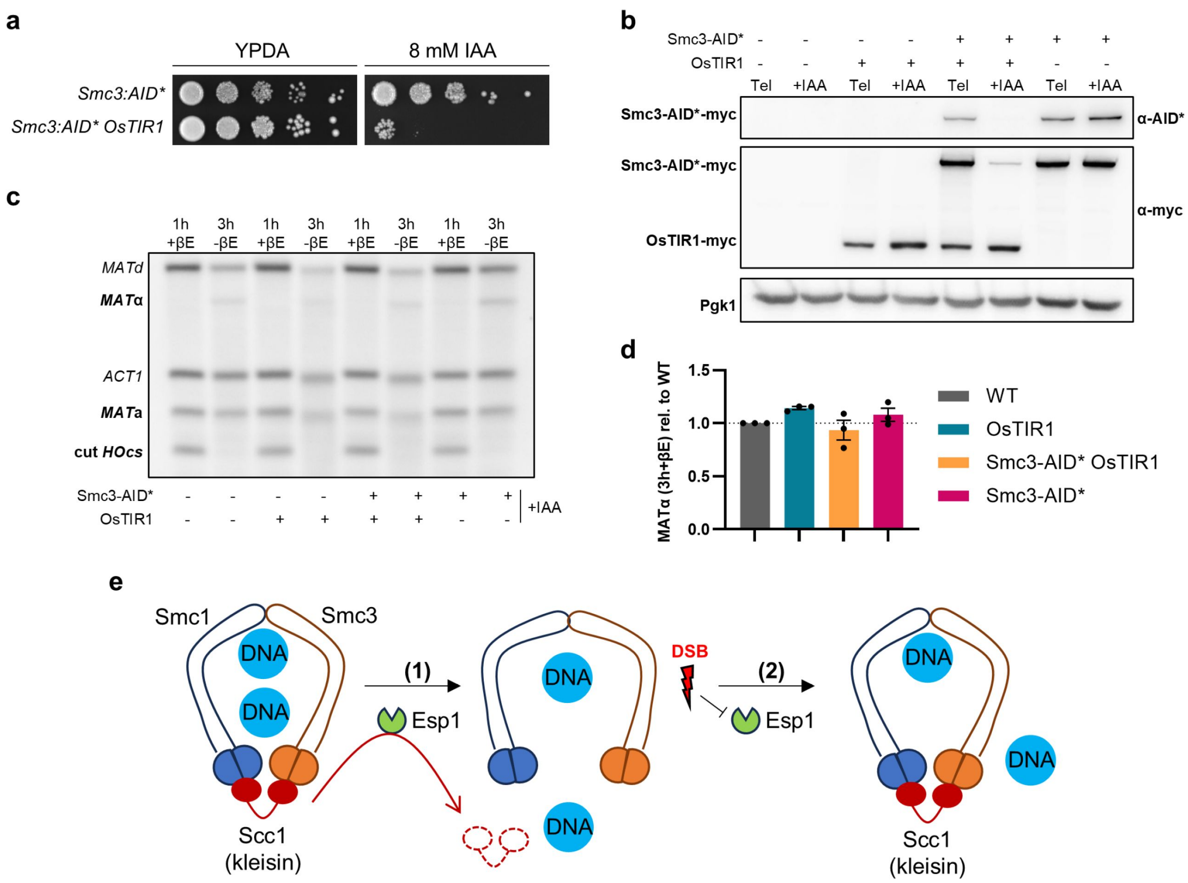


Figure 4.

Cohesin removal does not alter HR at the *MAT* locus in late mitosis.

(**a**) Serial dilution spot assay for the *smc3:aid** strains. The fitness effect of Smc3-aid* degradation was tested in the *smc3:aid* OsTIR1* strain in the presence of 8 mM IAA. The same strain without the ubiquitin-ligase *OsTIR1* was also used as the control. (**b-d**) Cells of four strains with all possible combination of the *smc3:aid** and *OsTIR1* pairs were treated as in **Figure 1a** but including a 1 h IAA step between the *cdc15-2* arrest and the induction of HO. Then, HO was induced for 1h with β-estradiol (1h+βE sample), after which it was washed off and the cells were allowed to recover from the DSB for 2h (3h-βE sample). IAA and the Tel arrest were maintained throughout the experiment. (**b**) Representative Western blot of the four strains at the time of the arrest (Tel) and 1h after IAA addition (+IAA). Note the confirmation of the different genotypic combinations, as well as the Smc3-aid* decline after IAA addition. (**c**) Representative Southern blot for the *MAT* switching assay of the four strains in the presence of 8 mM IAA. Note that HO cutting is less efficient than without IAA (see **Figures 3e** and **S6b,c**) but gene conversion is still possible after degrading Smc3-aid* in late mitosis. (**d**) Quantification of relative *MAT* switching (mean ± sem, n=3). Gene conversion to *MAT*α in the 3h-βE sample was normalized to the cut *HOcs* in the 1h+βE sample. Then, gene conversion yield was normalized to the wild type (no tagged Smc3, no OsTir1) strain. Tel: Telophase. +βE: β-Estradiol addition. −βE: β-Estradiol removal.+IAA: Indole-acetic acid. (**e**) Model of how DSBs reconstitute cohesin in late mitosis. The cohesin complex (Smc1-Smc3-Scc1) entraps sister chromatids before anaphase. (1) At anaphase onset, Esp1 (separase) is activated to cleave Scc1 and release sister chromatids for segregation. (2) If DSBs occur before cytokinesis, Scc1 returns and reconstitutes the cohesin complex. At least a fraction of the complex binds to chromatin. The reappearance of Scc1 likely occurs through the inhibition of Esp1 and/or post-translational protection of Scc1 by the DNA damage checkpoint.

not required for the HR-driven repair of the HO-DSB in telophase. Of note, it was previously reported that cohesin was not required for the *MAT* switching in G2/M-arrested cells, despite it being essential for efficient post-replicative DSB repair (Ünal et al., 2004).

Conclusions and perspectives

In this work we have addressed whether HR was molecularly efficient in yeast cells arrested in late mitosis. We have used the well-established *MAT* switching system, and found that HR is indeed working in late mitosis, complementing previous results that pointed in this direction both genetically and cytologically (Ayra-Plasencia and Machín, 2019). In addition, we have addressed the paradox of having HR in a cell mitotic stage with no sister chromatid cohesion. Our results suggest that residual cohesin, or at least residual chromatin-bound Smc1-Smc3 dimers, are complemented by incorporating *de novo* Scc1 (**Fig. 4e**). This fact suggests that chromosome structure may undergo significant changes after late DSBs. The nature of these changes remains unexplored and we cannot currently rule out that some kind of DI-cohesion could be established, perhaps globally. Alternatively, telophase chromatin may acquire new intramolecular loops as it occurs in G2/M-arrested cells (Piazza et al., 2021). Either of these may favour repair by HR with the intact sister or intramolecularly, although it seems still dispensable for *MAT* switching at this late cell cycle stage. Further research will be needed to confirm these hypotheses, as well as the whereabouts of cohesin and the consequences for late segregation after DSB repair.

Material and Methods

Strains and experimental conditions

The **table 1** contains the *S. cerevisiae* strains used in this work. Genetic backgrounds are either W303 or YPH499. Strain construction was performed by lithium acetate-based transformation on pre-made frozen competent cells (Knop et al., 1999).

Strains were cultured overnight in the YPD medium (10 $g{\cdot}L^{-1}$ yeast extract, 20 $g{\cdot}L^{-1}$ peptone and 20 $g{\cdot}L^{-1}$ glucose) with moderate shaking (180 rpm). The day after, 10-100 µL of grown cells were diluted into a flask containing an appropriate YPD volume and grown overnight at 25 ºC again. Finally, the exponentially growing culture was adjusted to OD_{600} = 0.5 to start the experiment.

In general, cells were first arrested in late mitosis by incubating the culture at 34 ºC for 3 hours (all strains carry the *cdc15-2* thermosensitive allele). Then, the culture was split in two, one subculture remained untreated (mock), and DSBs were generated in the second one. When a G2/M arrest was required, 15 $\mu g{\cdot}mL^{-1}$ nocodazole (Nz; Sigma-Aldrich, M1404) was added to the asynchronous culture, which was then incubated at 25°C for 3 h (with a 7.5 $\mu g{\cdot}mL^{-1}$ Nz shot at 2 h). To conditionally degrade the Smc3-aid* variant, 8 mM of the auxin 3-indol-acetic acid (IAA; Sigma-Aldrich, I2886) was added to the arrested cells 1 h prior to generating DSBs.

DSB generation was accomplished by incubating the culture for 1 to 3 hours with either 10 $\mu g{\cdot}mL^{-1}$ phleomycin (random DSBs) or 2 µM of β-estradiol (HO-driven DSBs). For the *HOcs* DSB, the corresponding strain harbours an integrative system based on a β-estradiol inducible promoter (Ottoz et al., 2014). These strains were constructed by transforming with the pRG464 plasmid.

Western blot for protein levels and phosphorylation

Western blotting was performed as reported before (Ayra-Plasencia and Machín, 2019). The trichloroacetic acid TCA method was used for protein extraction. Total protein was quantified in a Qubit 4 Fluorometer (Thermo Fisher Scientific, Q33227). Proteins were resolved in 7.5% SDS-PAGE gels and transferred to PVFD membranes (Pall Corporation, PVM020C099). The membrane was

Table 1.

Strains used in this work.

Strain	Relevant Genotype[1]	Origin	Figure
FM2344	(YPH499) *MATa ura3-52 lys2-801 ade2-101 trp1-Δ63 his3-Δ200 leu2-Δ1 bar1-Δ; cdc15-2:9myc::Hph; HMLα ∆hmr::HIS3MX*	Machín lab	(parental)
LSY4319-1C	(W303) *MATα leu2-3,112 trp1-1 can1-100 ura3-1 ade2-1 his3-11,15 RAD5 ∆hml ∆hmr*; *leu2-3::lexO4p:HO:FLAG:CYC1t::ACT1p:LexA-ER-B112-T:CYC1p::LEU2MX* (pRG646)	Symington lab	(parental)
FM2450	LSY4319-1C; *cdc15-2:9myc::Hph*	This work	3a,b
FM2520	FM2450; *SCC1:3myc::HIS3MX*	This work	1b,e
FM2531	FM2344; *leu2-3::lexO4p:HO:FLAG:CYC1t::ACT1p:LexA-ER-B112-T:CYC1p::LEU2MX*	This work	3d-f; 4b-d
FM2635	FM2450; *SMC1:6HA:HIS3MX*	This work	1c,e
FM2662	FM2531; *Δrad9::KanMX4*	This work	S5
FM2663	FM2531; *Δyku70::KanMX4*	This work	S5
FM2668	FM2531; *Δmre11::KanMX4*	This work	S5
FM2672	FM2531; *SMC3:AID*:9myc::KanMX*	This work	4a-d
FM2674	FM2450; *SMC3:3HA::HIS3MX*	This work	1d; 2c; S2; S3
FM2680	FM2672; *ura3-52::ADH1p:OsTIR1:9myc::URA3*	This work	4a-d; S6
FM3235	FM2520; *SMC1:6HA::NatNT2*	This work	2a,b; S1
FM3294	FM2531; *ura3-52::ADH1p:OsTIR1:9myc::URA3*	This work	4b-d

[1] Semicolons separate genetic modifications obtained through sequential transformation steps. Intermediate strains are omitted.

stained with Ponceau S solution (PanReac AppliChem, A2935) as a loading reference. The following primary antibodies were used for immunoblotting: mouse monoclonal α-HA (1:1,000; Sigma-Aldrich, H9658), mouse monoclonal α-myc (1: 5,000; Sigma-Aldrich, M4439), mouse monoclonal α-Pgk1 (1:5,000; Thermo Fisher Scientific, 22C5D8), mouse monoclonal α-miniaid (1:500; MBL, M214-3), mouse monoclonal α-Flag® (1:5000; Sigma-Aldrich, F3165), mouse monoclonal α-Rad53 (1:1000; Abcam, ab166859), and mouse monoclonal α-acSmc3 (1:5000; a gift from Katsuhiko Shirahige). The secondary antibody was a horseradish peroxidase polyclonal goat anti-mouse (from 1:5,000 to 1:10,000, depending on the primary antibody; Promega, W4021). Proteins were detected with the ECL chemiluminescence reagent (GE Healthcare, RPN2232), and visualized in a Vilber-Lourmat Fusion Solo S chamber.

Co-immunoprecipitation (co-IP) for the reassembly of the cohesin complex

For coI-P assays, approximately 100 OD_{600} units from the corresponding conditions were collected. Cell disruption was performed by resuspending the frozen pellets in 200 µL of Buffer A (50 mMHEPES-Na, 150 mMKCl, 1.5 mMMgCl $_2$, 0.5 mMDTT, 0.5%Triton X - 100, 1× EDTA-free protease inhibitor cocktail) together with glass beads. Samples were vortexed vigorously for 5 min, and additional lysis cycles were performed as needed, allowing 1 min incubation on ice between rounds. Lysates were transferred to a clean tube. 400 µL of Buffer A was used to wash the beads and pooled with the lysates. Samples were then centrifuged (12,000 rpm, 5 min, 4°C), and the clarified lysate s were transferred to new tubes. An aliquot of 70 µL was saved as whole cell extract (WCE) for western blot analysis. The remaining lysate was incubated with pre-equilibrated Pierce™ Anti-HA Magnetic Beads (Thermo Fisher Scientific; 88836) for 2 h at 4°C on a rotating wheel. After incubation, beads were placed on a magnetic rack, and washed five times with 200 µL of IPP150 buffer (10 mM Tris-Cl pH 7.5, 150 mMNaCl, 0.5%Triton X-100).

For elution, beads were resuspended in 25 µL of 1× SR buffer (8%SDS, 0.5 MTris - Cl pH 6.8) and incubated at 37 °C for 4 min. The supernatant was collected using a magnetic rack and transferred to a new tube containing 8.3 µL of 4× SS buffer (20%sucrose, 0.05% bromophenol blue, 0.1%sodiumazide) supplemented with β-mercaptoethanol. Samples were boiled for 2 min at 95°C, centrifuged (max speed, 30 se c) and analyzed by SDS-PAGE and western blotting.

Chromatin fractionation for the chromatin-bound cohesin

Biochemical fractionation of cells was performed as described before with little modifications (Cuevas-Bermúdez et al., 2020). Briefly, cells were collected and resuspended in 400 µl of buffer 1 (20 mM HEPES pH 8, 60 mM KCl, 15 mM NaCl, 10 mM $MgCl_2$, 1 mM $CaCl_2$, 0.8% Triton X-100, 1.25 M sucrose, 0.5 mM spermine). Breakage of cell walls was done by adding ~ 200 mg of glass beads and vortexing for 4 min. After centrifuging at 500 g for 5 min., supernatant was transfer to a new tube and 40 µl were kept as Input. Samples were centrifuge at 18000 g for 20 min, the supernatant was kept as S1, whereas the pellet was resuspended in 200 µl of Buffer 1 (50 µl were saved as P1). Samples were centrifuged again 18000 g for 20 min, the supernatant was kept as S2, and the pellet was resuspended in 200 µl of buffer 2 (20 mM HEPES pH 7.6, 45 mM NaCl, 7.5 mM $MgCl_2$, 20 mM EDTA pH 8, 10% glycerol, 1% NP-40, 2 M urea, 0.5 M sucrose, protease inhibitors) and 50 µl were saved as P2. Finally, samples were centrifuged one last time at the same conditions, keeping the supernatant as S3, and the pellet was resuspended in 50 µl of Laemli buffer, constituting the chromatin-enriched fraction (P3).

Chromatin immunoprecipitation (ChIP-qPCR) for the cohesin loading onto the HO DSB

For ChIP–qPCR, a total of 100 OD_{600} units of the treated cultures were collected and crosslinked with 1.4% formaldehyde for 15 min. Crosslinking was quenched with glycine (final concentration 125 mM) for 7 min. Cells were harvested by centrifugation (4000 rpm, 1 min), washed with PBS, transferred to screw-cap tubes, and frozen at –80°C.

Frozen pellets were resuspended in 300 µL of IP buffer (150 mMNaCl, 50 mMTris - HCl pH 7.5, 5 mMEDTA, 1%Triton X-100, 0.05% NP-40) supplemented with 1 mMPMSF and protease inhibitor cocktail (EDTA-free, Roche). Disruption was performed by bead beating with 500 µL of glass beads using a Bertin Precellys hoomgenizer for nine 20 s cycles at hard setting. Lysates were transferred to new tubes and supplemented with an additional 100 µL of IP buffer containing PMSF and protease inhibitors. Sam ples were clarified by centrifugation (15,000 rp,m10 min, 4 °C). Pellets were resuspended in 1 mL IP buffer with PMSF and protease inhibitors and sonicated in a Diagenode Bioruptor Plus (30 cycles: 30 s ON 30 s OFF, high power, 4°C). After sonication, samples were centrifuged (15,000 rpm, 10 min), and the supernatant was collected. A 200 µL aliquot was reserved as input. Input DNA was precipitated with 0.3 Msodiumacetate and 2.5 volumes of cold ethanol, centrifuged (15,000 rpm, 30 min), washed with 70%e thanol, and air-dried.

For imunoprecipitation, 400 µL of the sonicated chromatin was incubated with 40 µg of anti-HA antibody (Roche, 11666606001) in the Bioruptor at low power for 30 imn (30 s ON/30 s OFF cycles). Antibody-chromatin complexes were pelleted (13,000 rpm, 5 min), and the supernatant was incubated with 60 µL of Dynabeads Protein G (Invitrogen), pre-equilibrated in IP buffer, then incubated for 2 h at 4°C on a rotating wheel, and finally washed five times with IP buffer using a magnetic rack.

Both input and IP samples were resuspended in de-crosslinking buffer (1× TE, 1% SDS, 10 µg/mL RNase A, 1 mg/mL proteinase K) and incubated overnight at 65°C and purified using High Pure PCR product purification kit (Roche) for the qPCR.

qPCR for DSB resection

In the resection experiments, qPCR was performed on genomic DNA (gDNA) extracted by the glass beads/phenol method (Hoffman and Winston, 1987). Experimental details on the resection assay can be found in (Gnügge et al., 2018). Briefly, each gDNA sample was divided into two and one aliquot was digested with StyI-HF (NEB, R3500S). Then, qPCR reactions were mounted using the PowerUp™ SYBR™ Green Master Mix (Thermo Scientific, A25741) and run in a BioRad CFX384 Real-Time PCR instrument (10 µl final volume in 384-well block plates).

Resection was calculated as a normalized fraction ($f_{resected}$) to that of the HOcs effectively cut by HO (*f*). These two parameters were calculated by their corresponding formulas (Gnügge et al., 2018).

Southern blot for the MAT switching assay

In this case, gDNA was extracted by digesting with 50U lyticase (Sigma-Aldrich, L4025) a cell pellet previously resuspended in 200 µl of digestion buffer (1 % SDS, 100 mM NaCl, 50 mM Tris-HCl, and 10 mM EDTA). Then, the gDNA was phase separated by phenol:chloroform (PanReac AppliChem, A0944), precipitated with ethanol, resuspended in TE 1X with 10 µg·mL^{-1} RNase A, precipitated again, and resuspended in TE 1X. Then, the gDNA was digested with StyI, the restriction fragments separated on a 1.2% low EEOO LS Agarose gel, and finally Southern blotted. Southern blot was carried out by a saline downwards transference onto a positively charged nylon membrane (Hybond-N+, Amersham-GE; RPN303B) (García-Luis and Machín, 2014). DNA probes against *ACT1* and *MAT* loci were made using Fluorescein-12-dUTP Solution (ThermoFisher; R0101) and the Expand™ High Fidelity PCR System(Roche; 11732641001). Hybridization with fluorescein - labeled probes was performed overnight at 68°C. The following day, the membrane was incubated with an anti-fluorescein antibody coupled to alkaline phosphatase (Roche; 11426338910), and the chemiluminescent signal detected using CDP-star (Amersham; RPN3682). Blots were visualized in a Vilber-Lourmat Fusion Solo S chamber.

For band quantification, each individual band was normalized to the *ACT1* signal in the lane. Then, a second normalization to the *MAT*a band at the arrest was performed. To determine the yield of gene conversion to *MAT*α specifically, the *MAT*α band after recovery from the DSB (2-3 h after β-estradiol removal) was normalized to the amount of the cut *HOcs* band immediately before recovery (1h after β-estradiol addition).

Data representation and statistics

Three types of graphs were used to represent the data: bar charts, marker line graphs and box plots. In box plots, the center line represents the medians, box limits represent the 25th and 75th percentile, the whiskers extend to the 5th and 95th percentiles, and the dots represent outliers. Error bars in all bar and line charts represent the standard error of the mean (s.e.m.) of three independent biological replicates. Individual values are also represented as dots in the bar charts. Graphpad Prism 9 was used for generating the charts and for statistical analysis. Differences between experimental data points were generally estimated using the Mann-Whitney U test for box plots and one-way ANOVA with Tukey post hoc for bar charts.

Supplemental Figures

Figure S1.

Co-IP of the core cohesin copmlex.

This is a biological replicate of the experiment shown in **Figure 2a**.

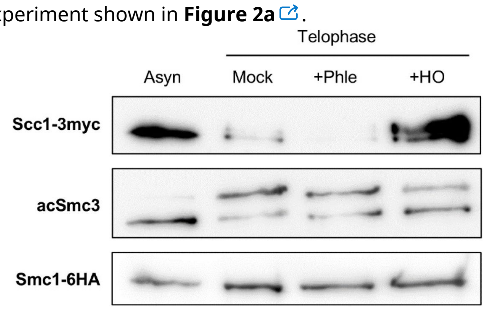


Figure S2.

Assessment of chromatin-bound cohesin after DSBs in late mitosis.

Related to **Figure 2b**. (**a**) Optimization of the chromatin fractionation using the chromatin-bound pool of Smc3 (acetylated Smc3; acSmc3) and the histone H3 (HH3). The best results were obtained after three successive rounds of fractionation (pellet #3, P3; see Material and Methods). S, supernatant; P, pellet. (**b**) The strain expressing Smc3-6HA was subjected to the same experiment and chromatin fractionation performed in **Figure 2b** for Smc1-6HA and Scc1-3myc. Note that there are more chromatin-bound acSmc3 after DSBs, especially after the HO DSB.

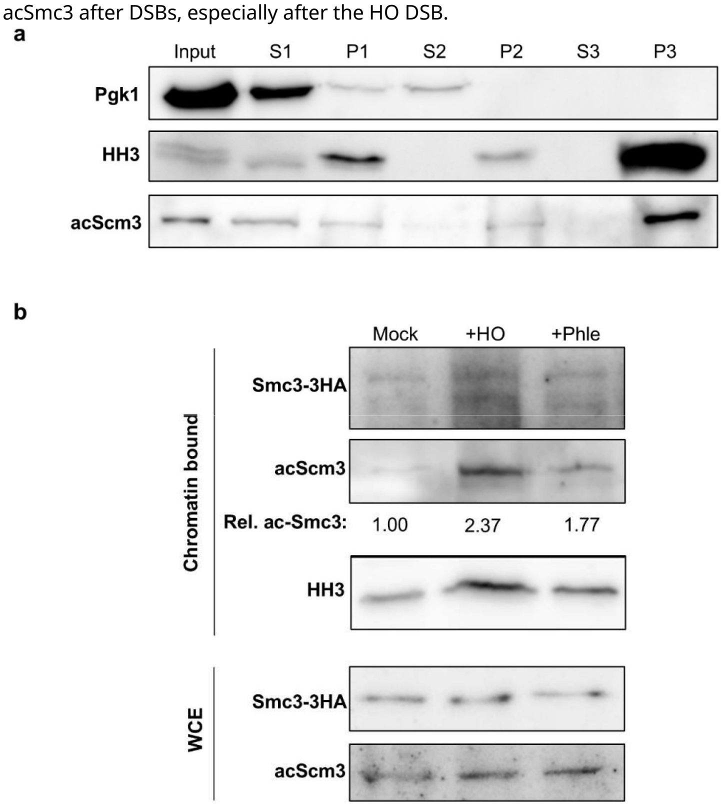


Figure S3.

Chromatin imumnoprecipitation (ChIP) of Smc3-6HA in G2/M.

Related to **Figure 2c**. The ChiP was performed upon cells arrested in G2/M with nocodazole and further subdivided into two subcultures, one in which the HO DSB was generated (+HO) and one without the DSB (-HO). The strain is unable to repair the break by gene conversion (Δ*hmr* Δ*hml*) and the HO induction lasted 2 h. Note the increase in Smc3 binding in the vicinity of *HOcs* after the DSB. This experiment was performed in parallel to the second repetition for Tel arrested cells shown in **Figure 2c**.

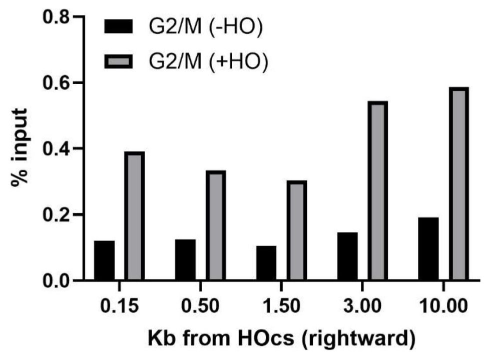
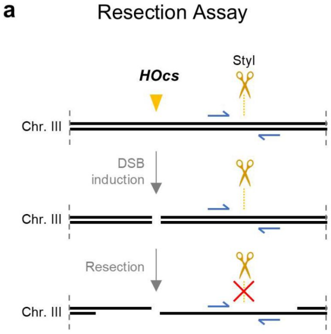


Figure S4.

The principle of the qPCR resection assay.

Related to **Figure 3a,b**. (**a**) Schematic representation of the HO resection assay. Primers (blue arrows) are designed to amplify a sequence that contains a *StyI* target site adjacent to the HO cutting site (*HOcs*). When this restriction enzyme is used on the extracted genomic DNA, amplification is inhibited. If resection extends beyond the *StyI* site, StyI does not cut and the primers can amplify. (**b**) Summary table of the amplification yield obtained after the StyI digestion during *HOcs* resection.

a



b qPCR amplicon

Undigested Template	StyI-digested Template
+++	-
+++	-
+	+

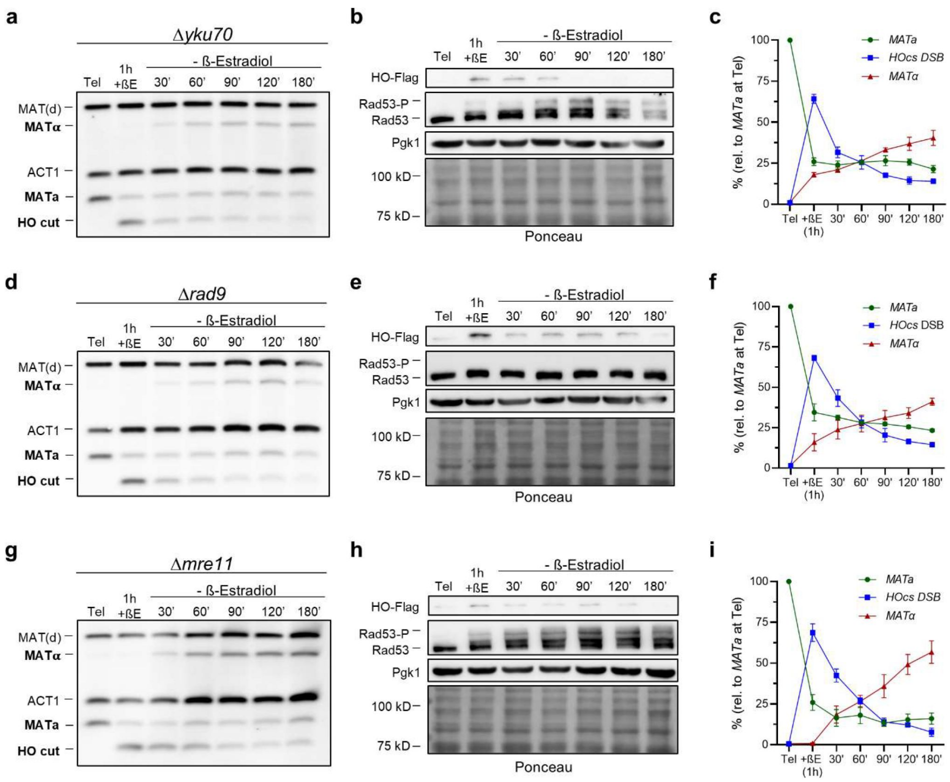


Figure S5.

Features of HR to repair the *HOcs* DSB in late mitosis.

Related to **Figure 3c-f**. Cells were first blocked in the *cdc15-2* arrest at 34 ºC for 3 hours (Tel). Then, the HO DSB was generated by adding β-estradiol. After 1 hour, the β-estradiol was washed away, and samples were taken to monitor the repair for 3 hours. (**a-c**) Kinetics of the *HOcs* DSB repair in *yku70*Δ. Yku70 is part of a conserved complex that recognizes a DSB and drives its repair towards NHEJ. (**d-f**) Kinetics of the *HOcs* DSB repair in *rad9*Δ. Rad9 is a mediator in the DDC that promotes activation of the effector kinase Rad53. (**g-i**) Kinetics of the *HOcs* DSB repair in *mre11* Δ. Mre1 is part of the MRX complex, which tethers the DSB ends and facilitates end resection. (**a**,**d**,**g**) Representative Western blot analyses for HO induction and subsequent degradation (tagged with Flag epitope) and the DSB sensing through Rad53 hyperphosphorylation (note that Rad53 does not get hyperphosphorylated in *rad9*Δ). A PGK1 western blot served as a housekeeping and Ponceau S staining of the membrane as a loading control for all lanes. The leftmost lane in the Ponceau S corresponds to the protein weight marker. (**b**,**e**,**h**) Representative Southern blots for the MAT switching assays in late mitosis. Alongside the MAT probe, a second probe against the *ACT1* gene (1.1 kb fragment) was included for normalization. The MAT probe also recognizes an allele-independent MAT distal fragment (2.2 kb). (**c**,**f**,**i**) Quantification of relative band intensities in the MAT switching Southern blots (mean ± sem, n=3). Individual values were normalized to the *ACT1* signals. Then, every lane was normalized to *MAT* a at the arrest. Note that repair kinetics is similar to that of the wild type in **Figure 3e,f**. Tel: Telophase. +βE: β-estradiol addition.

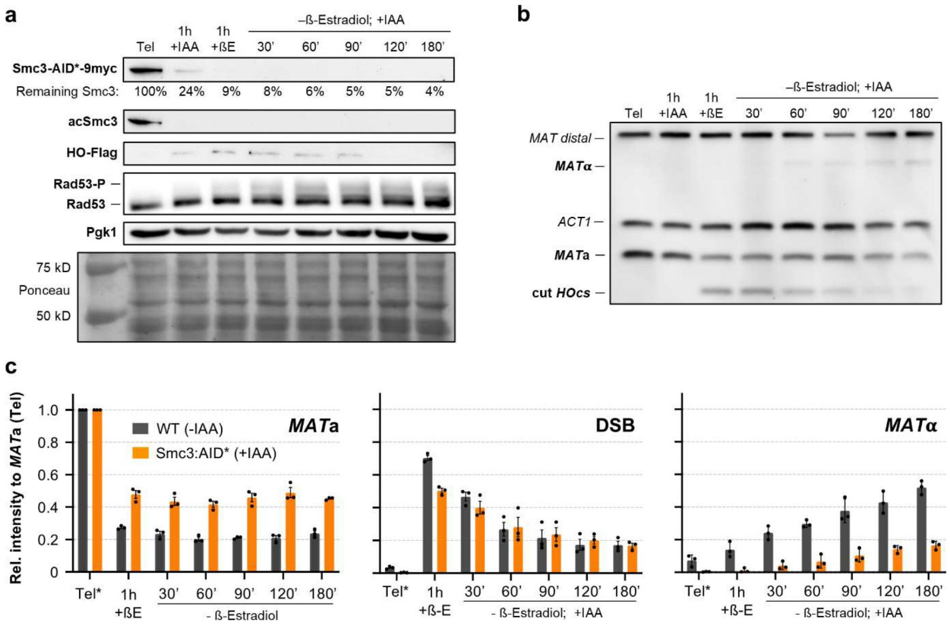


Figure S6.

MAT switching yield of Smc3-aid* OsTIR1 upon IAA addition.

Cells were treated as in **Figure 3d-f** but including a 1 h IAA step between the *cdc15-2* arrest and the induction of HO. IAA was then maintained throughout the experiment. (**a**) Representative Western blot as in **Figure 3d** but also including Smc3-aid* decline after IAA addition. Note that the chromatin-bound acSmc3 is degraded as well. (**b**) Representative Southern blot for the MAT switching assay as in **Figure 3e**. (**c**) Quantification of relative band intensities for MAT switching Southern blots (mean ± sem, n=3). Values were normalized as in **Figure 3f**. The MAT switching in the wild type strain without IAA is represented alongside for comparison. Note that there is a decrease in the cutting efficiency of HO (*MAT* a levels remained higher) as well as in the gene conversion into *MAT* α. However, these phenotypic changes are due to IAA rather than cohesin depletion (see **Figure 4c**). Tel: Telophase. +βE: β-Estradiol addition. +IAA: Indole-acetic acid.

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Additional information

Author contributions

J Ayra-Plasencia: methodology, investigation, formal analysis, visualization, curation, writing–original draft, writing–review and editing.

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E Hernández-Carralero: methodology, investigation, formal analysis, visualization writing– review and editing.

J García-Luis: methodology, investigation, formal analysis, visualization writing–review and editing.

L Symington: supervision, project administration, funding acquisition, writing–review and editing.

F Machín: conceptualization, visualization, resources, supervision, project administration, funding acquisition, curation, writing–original draft, writing–review and editing.

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Reviewer #1 (Public review):

Summary

The cohesin complex is essential for maintaining sister chromatid cohesion from S phase until anaphase. Beyond this canonical role, it is also recruited to double-strand breaks (DSBs), supporting both local and global post-replicative cohesion, a phenomenon first reported in 2004. In a previous study, Ayra-Plasencia et al. demonstrated that in telophase, DSBs can be repaired by homologous recombination (HR) through re-coalescence of sister chromatids (Ayra-Plasencia & Machín, 2019). In the present work, the authors provide further insights into DSB repair in late mitosis, showing that:

Scc1 is reloaded and reconstituted on chromatin together with Smc1.

HR occurs with high efficiency.

HR-driven MAT switching can occur in an Smc3-independent manner.

Strengths

The authors take full advantage of the yeast model system, employing the HO endonuclease to generate a single, site-specific DSB at the MAT locus on chromosome III. Combined with careful cell synchronization, this setup allows them to monitor HR-mediated repair events specifically in G2/M and late mitosis. Their demonstration that full-length Scc1 can be recovered upon DSB induction is compelling. Most importantly, the finding that efficient HR can take place during M phase is significant, as HR has long been thought to be largely inhibited at this stage of the cell cycle.

Weaknesses

While the authors provide evidence for Scc1 recovery and efficient HR in late mitosis, some critical points need to be clarified to improve the impact and interpretability of the study.

https://doi.org/10.7554/eLife.92706.2.sa2

Reviewer #2 (Public review):

Cohesin drive inter-sister repair of DNA breaks by homologous recombination (HR) in G2/M. Cohesion is lost at the metaphase to anaphase transition upon digestion of the Scc1 subunit of cohesin by Esp1, raising the question as to whether and how break repair by HR could occur in late mitosis (late-M).

Here the author investigate the behavior of cohesin in cells arrested in telophase and experiencing a DNA break at the mating-type locus on chr. III (a specialized recombination process required for mating-type switching) or upon random DNA break formation with the drug phleomycin.

The revised version of the manuscript now convincingly establishes three facts:

- The cohesin subunit Scc1 can re-associate with chromatin and the other Smc1-3 subunits upon formation of an unrepairable DSB at MAT in telophase.
- HR can occur in telophase-arrested cells
- Cohesin (an a fortiori cohesin that reassociated with chromatin) plays no role in non-allelic HR in telophase in the specific context of MAT switching.

Unfortunately, the role of cohesin re-association with chromatin for the allelic inter-sister repair by HR is not addressed. In the absence of such evidence, the main claims of the paper making up the title (cohesin re-association and HR repair) appear disconnected. Even if the very last sentence of the abstract corrects the false sense from the title and the rest of the abstract that cohesin reconstitution has somehow something to do with efficient HR in late mitosis, I think a general rewriting of the abstract and a different title would better lift any ambiguity about the conclusions of the paper.

https://doi.org/10.7554/eLife.92706.2.sa1

Author response:

The following is the authors' response to the original reviews

We would like to thank the reviewers for taking the time to thoroughly revise our work. We have considered their suggestions carefully and tried our best to respond to them point by point. Based on their recommendations, two major issues came forward: (1) the strength of our claims about the involvement of cohesin in HR-driven repair in late mitosis; and (2) the underlying mechanism that reconstitutes cohesin in late mitosis after DNA damage. In this revision, we focused on the former and left the latter out (yet it is discussed). We considered that the question of how cohesin returns in late mitosis after DNA damage is important and worthy of further research, but it is beyond the scope of this study (as it is the putative role of condensin). Thus, we have focused on buttressing our main claims, as otherwise pointed out by the reviewers. What have we done to strengthen the role of cohesin in late mitotic DSB repair?

(1) We have biologically replicated and quantified the reappearance of Scc1 after DSB generation (new Figure 1e). We have also quantified changes for the other core subunits (new Figure 1c-e).

(2) We now show that the newly synthetized Scc1 serves to assemble back the cohesin complex (new Figure 2a and S1).

(3) We have performed chromatin fractionation and show that cohesin binding to chromatin increases after the HO-induced DSB (new Figure 2b and S2).

(4) We have performed ChIP assays and show that, despite the increase in the chromatin-bound fraction, the *HOcs* DSB does not recruit new cohesin to the locus (new Figure 2c and S3).

(5) A key assertion in the preprint version was that depleting cohesin using the auxin degron system impairs HR-driven MAT switching. This claim was based on a direct comparison of cultures treated or not with auxin (-/+ IAA). However, during the revision process, we

realized that auxin treatment itself could interfere with MAT switching. Firstly, we noticed a diminished *HOcs* cutting efficiency by HO in +IAA cultures (Figure S6). Secondly, the apparently dramatic delay in gene conversion to *MAT*α could actually be related to other undesirable effects of IAA downstream in the repair process. Thus, we decided to repeat this experiment with strains that differ in their response to auxin, so that we could compare all strains in the presence of auxin. We compared four isogenic strains: *SMC3*; *SMC3-aid**; *SMC3* + *OsTIR1*; and *SMC3-aid** + *OsTIR1*. As a result, we can now show that cohesin depletion does not affect MAT switching (see new Figure 4b-d).

(6) We recently reported a negative chemical interaction between auxin and phleomycin. Auxin appears to diminish the ability of phleomycin to generate DSBs (Comm Biol 2025, doi: 10.1038/s42003-025-08416-x; see Figures S14 and S15 in that paper). While the underlying nature of this interaction is unknown to us (we are working on it), this leads us to omit the coalescence assay included in the preprint version (old Figure 4c), as the diminished coalescence upon IAA addition is actually due to this effect rather than cohesin depletion. This is also in agreement with the new data we include in the revised version, in which we observed only minor changes in cohesin reconstitution and chromatin binding after phleomycin (Figure 2a,b; S1 and S2).

(7) In addition to addressing these reviewers' requests, we have better characterized the *MAT* switching in late mitosis by incorporating the kinetics of *rad9*Δ (deficient in the DNA damage checkpoint), *yku70*Δ (deficient in non-homologous end joining) and *mre11*Δ (deficient in DSB end tethering). The effect of *rad52*Δ (deficient in HR) has been described elsewhere (our iScience 2024, 10.1016/j.isci.2024.110250).

As a result of these new experiments, new figure panels have been added in the main figures and as supplementary figures. To make room for the these panels in the main figures and keep the short report format, the following changes have been made: (i) old figures and new panels have been combined into four main figures, (ii) some panels from the old figures have been moved to supplementary figures, and (iii) some panels have been reordered for the sake of simplicity and fluidity in the main text.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *The cohesin complex maintains sister chromatid cohesion from S phase to anaphase. Beyond that, DSBs trigger cohesin recruitment and post-replication cohesion at both damage sites and globally, which was originally reported in 2004. In their recent study, Ayra-Plasencia et al reported in telophase, DSBs are repaired via HR with re-coalesced sister chromatids (Ayra-Plasencia & Machín, 2019). In this study, they show that HR occurs in a Smc3-dependent way in late mitosis.*
>
> *Strengths:*
>
> *The authors take great advantage of the yeast system, they check the DSB processing and repair of a single DSB generated by HO endonuclease, which cuts the MAT locus in chromosome III. In combination with cell synchronization, they detect the HR repair during G2/M or late mitosis. and the cohesin subunit SMC3 is critical for this repair. Beyond that, full-length Scc1 protein can be recovered upon DSBs.*
>
> *Weaknesses:*
>
> *These new results basically support their proposal although with a very limited molecular mechanistic progression, especially compared with their recent work.*

eLife

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *The manuscript "Cohesin still drives homologous recombination repair of DNA double-strand breaks in late mitosis" by Ayra-Plasencia et al. investigates regulations of HR repair in conditional cdc15 mutants, which arrests the cell cycle in late anaphase/telophase. Using a non-competitive MAT switching system of S. cerevisiae, they show that a DSB in telophase-arrested cells elicits a delayed DNA damage checkpoint response and resection. Using a degron allele of SMC3 they show that MATa-to-alpha switching requires cohesin in this context. The presence of a DSB in telophase-arrested cells leads to an increase in the kleisin subunit Scc1 and a partial rejoining of sister chromatids after they have separated in a subset of cells.*
>
> *Strengths:*
>
> *The experiments presented are well-controlled. The induction systems are clean and well thought-out.*
>
> *Weaknesses:*
>
> *The manuscript is very preliminary, and I have reservations about its physiological relevance. I also have reservations regarding the usage of MAT to make the point that inter-sister repair can occur in late mitosis.*

Regarding these two weaknesses:

- Physiological relevance: This is something we already addressed in our previous research work (Nat Commun. 2019; 10(1):2862. doi: 10.1038/s41467-019-10742-8), and which was further discussed in a follow-up theoretical paper (Bioessays. 2020 ;42(7):e2000021. doi: 10.1002/bies.202000021). In summary, this is physiologically relevant because a DSB in anaphase activates a late-mitotic checkpoint so the DSB can be repaired before cytokinesis. The fact that anaphase is quick and only a minor fraction of cells get a DSB in this cell cycle stage in an asynchronous population does not preclude its importance since it is enough a single mis-repaired DSB in hundreds of cells to mutate a population in an health- or evolution-relevant way.

- MAT system in late mitosis: It was not our intention to use the MAT switching assay to state that inter-sister repair can occur in late-M. The purpose was to address whether HR was fully functional in this non-G2/M non-G1 stage. Having said that, it is very challenging to design a strategy based on sequence-specific DSB to tackle the inter-sister repair in late-M. Any endonuclease-generated DSB is going to cut in both sisters. This is something we also deeply discussed in our previous works (Nat Commun & Bioessays).

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations For The Authors):***
>
> *Major points:*
>
> *(1) Smc3 degradation affects Rad53 activation upon DSBs, and this may directly lead to HR repair deficiency. Smc3 also could be phosphorylated by ATM and functions in DNA damage checkpoint activation, these alternative possibilities should also be tested before addressing the bona fide role of Smc3 in this context.*

Our previous data already suggested that Rad53 hyperphosphorylation still occurs after Smc3 degradation (Figure S6). Regardless, the question of whether the DNA damage checkpoint

(DDC) may play a distinct role in the MAT switching has been addressed in this revision by comparing *RAD9* versus *rad9*Δ. Rad9 is a mediator in the DDC required for the activation of Rad53. We have seen that MAT switching in *rad9*Δ is as efficient as in *RAD9* (new Figure S5d-f).

On the other hand, our new results, in which we have compared four different strains with all auxin system combinations in the presence of auxin, show that cohesin depletion does not affect MAT switching. Previously, we compared minus versus plus auxin and noticed diminished HO cutting efficiency. Thus, we repeated this experiment with four isogenic strains (*SMC3*; *SMC3-aid**; *SMC3 + OsTIR1*; and *SMC3-aid* + OsTIR1*) that differ in their response to auxin and ability to degrade cohesin, so that we could compare all strains in the presence of auxin. As a result, we can now affirm that cohesin depletion does not affect MAT switching (see new Figure 4b-d). Therefore, HR appears efficient after cohesin depletion.

> *(2) The requirement of cohesin subunit Smc3 and "coincidently" recovery of Scc1 are not sufficient to claim they act as a cohesin complex in this scenario. CoIP in the chromatin fraction after DSBs to prove the cohesin complex formation is recommended. If they act as a complex, are cohesin loader Scc2/4 required?*

We have constructed a *SMC3-HA SCC1-myc* strain. We have purified the chromatin-bound fraction as well as performing the co-IP. We have found Smc1-acSmc3-Scc1 forms a complex after Scc1 returns, and that at least a fraction of this complex binds to the chromatin in our HO model of DSBs in late anaphase (the *cdc15-2* arrest). This is now shown in the new Figures 2a,b and S1,S2.

As for the requirement of Scc2/4, we consider that the mechanisms underlying how Scc1 comes back, how a new cohesin complex is reassembled, and how it can partly bind to the chromatin in late anaphase are beyond the scope of this study and worth pursuing in a follow-up story.

> *(3) Figure 3b. acetylated SMC3 was prominently detected in the absence of DSBs. During the cohesion cycle, the cohesin was released from chromatin in a separase-dependent manner at the anaphase onset. Released Smc3 was deacetylated by Hos1 subsequently. In principle, the acSMC3 level could be very low in late mitosis.*

In that figure (now renumbered as Fig S6), we did detect acetylated Smc3 for the remnant Smc3 still found in late mitosis, however, a direct comparison between the acetylated versus non-acetylated pools was not performed, and would require more sophisticated approaches. Note that blots are distinctly exposed until the band is detected, and that signal intensity is antibody-specific. The presence of an acSmc3 pool in the *cdc15-2* arrest is now further confirmed by the new blots in Figures 2a, S1 and S2b.

On the other hand, previous time course experiments from G1 and G2/M releases point out that Smc3 deacetylation is incomplete in anaphase, with up to 30% of acetylated Smc3 remaining (Beckouët et al, 2010 doi:10.1016/j.molcel.2010.08.008). This is consistent with the presence of acSmc3 in the *cdc15-2* arrest.

> *(4) Did the author examine the acSMC3 levels returning after DSB, as Scc1's levels? If so, how about the Eco1's protein level? Chromatin fractionation could be conducted to check the chromatin-bound SMC3, acSMC3/Eco1, SCC1, SCC1 phosphorylation, and SMC1. These results will tell us whether cohesin functions in DSB repair in late M in a cohesion state.*

As stated above, we have now determined that cohesin depletion does not affect HR-driven MAT switching. As for the other questions, yes, we have performed both an assessment of acSmc3 in the pull down and chromatin fractionation, before and after DSBs (new Figures 2a, S1 and S2b). Interestingly, we have noticed a difference between the HO-generated and the phle-generated DSBs. It appears that the former leads to a better reconstituted Smc1-acSmc3-Scc1 complex and more chromatin-bound cohesin. The overall acSmc3 levels do not appear to significantly change in the whole cell extracts, although there could be further posttranslational modifications in telophase (see the changes in intensity between the two acSmc3 bands in Figure S1).

The role of Eco1 has not been directly addressed but is discussed. The main point here is that Eco1 levels may be low after G2/M (e.g., Lyons and Morgan, 2011), but there is still a significant acSmc3 pool in anaphase as Hof1 does not deacetylate all Smc3 (Beckouët et al., 2010).

> *(5) Figure 4a, the return of full-length Scc1 is based on a single experiment. What's the mechanism? Inhibition of cleavage or re-expression? How about its mRNA levels?*

We have repeated the full-length Scc1 experiment two more times. Now, an expression graph is included as a new Figure 1e. The two other subunits, Smc1 and Smc3, have been assessed as well, with no major changes in abundance (new Figure 1c and d).

We feel that the exact molecular mechanism of how Scc1 returns is beyond the scope of this study, but we discuss that the DDC may either inactivate separase or protect Scc1 against it. Indeed, there is literature that supports both mechanisms (e.g., Heidinger-Pauli et al., 2008 doi:10.1016/j.molcel.2008.06.005; Yam et al., 2020 doi:10.1093/nar/gkaa355).

> *Minor points:*
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> *(6) FACS data should be shown for all cell synchronization experiments.*

From our previous own works, FACS profiles add little to late-M experiments. To properly confirm late-M, microscopy is a must. FACS cannot differentiate between G2/M (metaphase-like), anaphase, telophase and the ensuing G1 (as *cdc15-2* cells do not immediately split apart after re-entering G1). In all experiments, Tel samples (late-M *cdc15-2* arrest) were characterized by >95% large budded binucleated cells.

> *(7) Figure 1d, A loading control of Rad53-P in is missing. The "Arrest" samples should be loaded again on the right to confirm the shift of Rad53, but not due to "smiling gels".*

It is true that the blot on the right has a right-handed smile; however, it is very clear the presence of the Rad53/Rad53-P partner. Because there is not a full shift from Rad53 to Rad53-P, the concern of misidentifying Rad53-P as a result of a blot smile is unfounded.

> *(8) Figure 1c, After the HO cut, the resected DNA at the 726 bp site reaches to platform at about 4 hrs, while it still increases at the 5.6 kb site. Thus, it is difficult to conclude that "The time to reach half of the maximum possible resection (t1/2) was ~1 h at 0.7 Kb and ~2.5 h at 5.7 Kb from the DSB, respectively".*

We assumed that both loci reach the plateau at 0.8 (which is consistent with other studies), so the t1/2 was calculated when the resected intersected 0.4.

> *(9) Figure 2b and 2c are wrongly labeled.*

We have fixed this (now Fig. 3d and e).

> *(10) Figure 2d, Double check and make sure the quantitative data reflects the representative result. E.g. in Figure 2b (in fact should be 2c). For instance, in Figure 2b, the MATa signals seem to remain stable from 60' to 180', but they keep increasing in Figure 2d. In Yamaguchi & James E. Haber's paper, the signals and changes of MATa and MATa over time are way stronger compared to this study.*

We have double checked this. It is true that the sum of *MAT*α, *MAT*alpha and cut *HOcs* bands throughout the assay does not have the intensity seen for MATa before the HO induction (Tel), but *MAT*alpha and *HOcs* signals cannot be established based on the equimolarity of the reaction as all band signals are probe-specific (the best indication of this can be seen in the signal comparison between *MAT*α and *MAT* distal at Tel). Alternatively, some resected *HOcs* may remain unrepaired.

As for the referred example (now Figure 3e), note that they are double normalized to *ACT1* and *MAT*α (Tel), and the *ACT1* band gets fainter after 60'. This explains the increase in the MATalpha quantification in spite of what is apparently seen in the blot.

> *(11) Typos and fonts: e.g. lines 111-112; line 76 "his link".*

We have fixed this. Thanks.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *Major concerns:*
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> *(1) Physiological relevance. The authors show that HR can happen in the anaphase to telophase interval, yet does it outside of an hours-long artificial arrest upon inactivation of Cdc15? It is this reviewer's understanding that the duration of the anaphase to telophase transition is short, in the order of minutes. In fact, break signaling and resection are delayed by ~1 hour (Fig. 1), which suggests that cells avoid dealing with the damage and engaging in HR in the anaphase-telophase interval. Is there any described physiological context or checkpoint that blocks this transition for extended periods, that would make any of the findings in this paper relevant?*

This concern about the physiological relevance was addressed in our previous study (Nat Commun. 2019; 10(1):2862. doi: 10.1038/s41467-019-10742-8). In that paper's Figure 1, we showed that G1 re-entry after a *cdc15-2* release was delayed by several hours when DSBs had been previously generated at the *cdc15-2* arrest. We also showed that such a delay depended on Rad9 (i.e., the DNA damage checkpoint). In addition, synchronized (not arrested) cells transiting through anaphase responded to DSB generation by slowing anaphase transition while partly regressing chromosome segregation (Figure S7 in that paper).

> *(2) Methodological caveats. It is unclear why the authors chose to study DSB-repair in the context of MATa-to-alpha switching (which uses an ectopic donor on the other chromosome arm) as a model for inter-sister repair. It creates a disconnect in the claims of the paper, which means to study inter-sister repair. Studying the kinetics of DSB repair by cytology following low-dose irradiation or radiomimetic drugs would have been a better option. Phleomycin is used in Fig. 4, but the repair kinetics (e.g. Rad52 foci) is not studied.*

The MAT switching assay was used here to address how much HR was functional in late-M compared to G2/M (metaphase-like). Then, it was employed to check how cohesin depletion hampers HR in late-M. Even though this is something we already deeply discussed previously (Nat Commun. 2019; 10(1):2862. doi: 10.1038/s41467-019-10742-8; Bioessays. 2020

;42(7):e2000021. doi: 10.1002/bies.202000021), it is worth recapitulating the methodological challenges that the study of inter-sister repair has in late-M: (i) endonuclease-based DSBs are going to generate two DSBs, one per sister chromatid; (ii) the use of a homologous chromosome without the cutting site as a template is pointless because a sister of the homolog is always going to co-segregate with the broken chromatid, and the same caveat applies for any other ectopic sequence. In this context, the *MAT*a with the *HML* ectopic intrachromosomal sequence is as valid as any other option, with the advantage that it is a very well-known system.

On the other hand, most of the reviewer's concerns about the inter-sister repair by cytology and the role of Rad52 was addressed in our previous paper (Nat Commun). Note that our new results about the cohesin role on MAT switching show that this HR-mediated DSB repair does not depend on cohesin (new Figure 4b-d).

> *(3) Preliminary work. The requirement of cohesin for MAT switching in cdc15 mutants would have warranted several additional experiments. Indeed, Cohesin has been shown to regulate homology search in multiple ways upon DNA damage checkpoint-induced metaphase-arrest (see Piazza et al. Nat Cell Biol 2021 (10.1038/s41556-021-00783-x), not cited in the current manuscript). Consequently, is the effect of cohesin observed in the MAT system specific to telophase or is it true in other cell-cycle phases? What is the mechanism behind this requirement (one may expect it not to depend on the sister since the HML donor is available within the damaged chromatid)? Does cohesin re-accumulate around the DSB site or genome-wide? How does the Esp1 activity decay from anaphase onset? Is cohesin required for the horseshoe folding of chr. III involved in MATa-to-alpha switching? Furthermore, condensin is involved in MATa-specific switching (Li et al. PLoS Genet 2019, 10.1371/journal.pgen.1008339), and condensin remains active on chromatin in cdc15 arrested cells, as shown on chr. XII (Lazar-Stefanita et al. EMBO J. 2017 10.15252/embj.201797342), which calls for determining the impact contribution of condensin in the recoil of the right ch.XII arm (Fig 4c) and on MAT switching.*

There are several points here:

> *- Is the effect of cohesin observed in the MAT system specific to telophase or is it true in other cell-cycle phases?*

Our new results show that cohesin depletion does not affect MAT switching when four different strains with all auxin system combinations are compared in the presence of auxin. Previously, when we compared minus versus plus auxin, we noticed diminished HO cutting efficiency. Therefore, we repeated the experiment using four isogenic strains (*SMC3*, *SMC3-aid**, *SMC3* + *OsTIR1*, and *SMC3-aid** + *OsTIR1*), which differ in their response to auxin and ability to degrade cohesin. This allowed us to compare all strains in the presence of auxin. As a result, we can now confirm that cohesin depletion does not affect MAT switching (see the new Figures 4b–d). Therefore, HR appears efficient after cohesin depletion. In agreement, the new ChIPs we have performed do not detect an increment in local cohesin after the HO DSB in telophase (but it does in cells arrested in G2/M).

> *- What is the mechanism behind this requirement (one may expect it not to depend on the sister since the HML donor is available within the damaged chromatid)?*

As just said, we have changed our previous conclusion on cohesin and MAT switching. It was an effect of auxin addition rather than cohesin depletion.

> *- Does cohesin re-accumulate around the DSB site or genome-wide?*

We have performed ChIP around the *HOcs*. We have found that it does accumulate in G2/M after HO induction, but it does not in telophase (new Figures 2c and S3). As for the global binding of cohesin, our chromatin fractionation data suggest there is ~2-fold increase in Smc1-Smc3, which also binds to the newly formed Scc1, rendering an overall increase in the chromatin-bound canonical complex (new Figures 2b and S2). Altogether, this suggests a genome-wide binding but with little role in the repair of HO DSBs.

> *- How does the Esp1 activity decay from anaphase onset?*

We have not checked this here but it is an interesting question for a follow-up story.

> *- Is cohesin required for the horseshoe folding of chr. III involved in MATa-to-alpha switching?*

Probably not in view of our new data in Figures 2c and 4b-d. The Piazza papers are cited and discussed.

> *- Contribution of condensin in the recoil of the right ch.XII arm (Fig 4c) and on MAT switching.*

The role of condensin, which overtakes some cohesin function in late-M as the reviewer reminds, is worth studying indeed. However, we feel this deserves a separate and focus-on study. We does discuss, though, that condensin loading onto the arms in anaphase may prevent Smc1-Smc3 from loading after DSBs.

> *Other points:*
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> *(4) Is the retrograde behavior in Fig. 4c dependent on recombination?*

No, this is something we addressed in our previous paper (see Figure 4 in Nat Commun. 2019; 10(1):2862. doi: 10.1038/s41467-019-10742-8).

> *(5) Fig 3c: add a scheme of the system.*

A scheme was already shown in the old Figure 2a (note that the old Fig 3c is now Fig S6).

> *(6) Fig 3b: annotate as in Fig 2b.*

We have fixed this (now the referred figures are S6a and 3d, respectively).

> *(7) Authors used IAA concentrations 4- to 8-fold higher than commonly used. Given the solubility of IAA in DMSO (the most commonly used solvent), it is likely that authors treated their cells with >2% DMSO. This is expected to have broad transcriptional and physiological effects on yeast. A comparison of +IAA samples with a mock (DMSO) treatment would be more appropriate than a lack of treatment.*

The IAA stock solution was 500 mM in DMSO, so the final DMSO concentration for an 8 mM IAA solution was 1.6% (v/v). Although the stock concentration was high and some precipitation was observed during preparation, we always heated, sonicated, and vigorously vortexed the stock tube before adding IAA to the cultures. Thus, we kept the uncertainty in the final IAA concentration to a minimum.

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